SPLITTING DISTINGUISHED TRIANGLES VIA THE LONG EXACT HOM SEQUENCE

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 929-944, following the statement at 916-927.

Editorial note. No omissions: all three parts are executed; the Yoneda identification is named.

Allowed standard prerequisites (editorial).

- Representable functors are homological. lemma-representable-homological (carded L26235, cited at ~929). Exactness of the Hom sequences in (1) and (2).
- Direct sums of distinguished triangles. lemma-direct-sum-triangles (carded L26186, cited at ~943). Assembles the triangle in (3).

Source reference. Original lemma 05QS: sources/stacks-derived/pinned-original-source.tex, source lines 856–856. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0149: sources/stacks-derived/pinned-original-source.tex, source lines 603–603. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{D}$ be a pre-triangulated category. Let (X, Y, Z, f, g, h) be a distinguished triangle.*

(1) *If $h = 0$, then there exists a right inverse $s : Z \to Y$ to g.*
(2) *For any right inverse $s : Z \to Y$ of g the map $f \oplus s : X \oplus Z \to Y$ is an isomorphism.*
(3) *For any objects X', Z' of $\mathcal{D}$ the triangle $(X', X' \oplus Z', Z', (1,0), (0,1), 0)$ is distinguished.*

Proof. To see (1) use that $\mathrm{Hom}_{\mathcal{D}}(Z, Y) \to \mathrm{Hom}_{\mathcal{D}}(Z, Z) \to \mathrm{Hom}_{\mathcal{D}}(Z, X[1])$ is exact by Lemma 0149. By the same token, if s is as in (2), then $h = 0$ and the sequence

$$0 \to \mathrm{Hom}_{\mathcal{D}}(W, X) \to \mathrm{Hom}_{\mathcal{D}}(W, Y) \to \mathrm{Hom}_{\mathcal{D}}(W, Z) \to 0$$

is split exact (split by $s : Z \to Y$). Hence by Yoneda's lemma we see that $X \oplus Z \to Y$ is an isomorphism. The last assertion follows from TR1 and Lemma 05QS. □

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.